LESSON 6.15

Simplify Algebraic Expressions

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can simplify algebraic expressions by combining like terms.

WHY THIS MATTERS The music studio is placing supply orders for three rehearsal rooms. Room A needs 4 microphones and 3 stands. Room B needs 2 microphones and 5 stands. Room C needs 6 microphones and 1 stand. Instead of ordering separately, you can combine: (4 + 2 + 6) microphones and (3 + 5 + 1) stands = 12 microphones and 9 stands. That's simplifying by combining like items!

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Like Terms *Spanish: Términos semejantes*	Terms with the same variable raised to the same exponent, like 2x and 5x.	5x and 3x are like terms (both x); 5x and 5y are NOT (different variables); 2x and 2x² are NOT (different powers)
Combine *Spanish: Combinar*	To add or subtract terms with the same variable and exponent into one.	8m + 3m = 11m — add the coefficients (8 + 3 = 11) and keep the variable (m)
Simplify *Spanish: Simplificar*	To rewrite an expression in a shorter form that is still equal to the original.	2a + 5a + 3 simplifies to 7a + 3 — the two a-terms combine, the constant stays
Coefficient *Spanish: Coeficiente*	The number in front of a letter, like the 3 in 3x.	In 9x, the coefficient is 9 — it tells you 'nine groups of x'; if x = 2, then 9x = 18
Term *Spanish: Término*	A single number, variable, or product of them, separated by + or − signs.	In 3x + 5 − 2x, the terms are 3x, 5, and 2x.
Distributive property *Spanish: Propiedad distributiva*	Multiplying a number by everything inside the parentheses: a(b + c) = ab + ac.	3(x + 4) = 3·x + 3·4 = 3x + 12.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

The rooms need microphones and stands: (4 + 2 + 6) microphones and (3 + 5 + 1) stands.

1. Microphones are like items, so I add: 4 + 2 + 6 = 12 microphones.
2. Stands are like items, so I add: 3 + 5 + 1 = 9 stands.
3. In algebra this is the same as 4m + 2m + 6m + 3s + 5s + 1s = 12m + 9s.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Let's simplify 5x + 3x. Are these like terms?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- Two terms are like terms when they have the same _____. You cannot combine _____ and _____ because they represent different _____.

Word bank Like Terms | Combine | Simplify | Coefficient | Term | Distributive property | variable | like terms | coefficient | combine | same variable | different

WATCH OUT!

A common mistake

A common mistake in Simplify Algebraic Expressions is sweeping a constant into the variable terms

LESSON 6.15

Simplify Algebraic Expressions

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each pair of terms by whether they are like terms (can be combined) or not.

Groups: Like Terms | Not Like Terms

_____ 4m and 7m

_____ 3x and 3y

_____ 5n and n

_____ 8 and 2p

_____ 6a and 9a

_____ $2x^2$ and 2x

HINT Like terms match in BOTH the variable letter and its exponent. Plain numbers only pair with plain numbers.

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

Simplify 7x + 3x.

(A) 10x

(B) $10x^2$

(C) 21x

(D) 73x

HINT 7x and 3x have the same variable x, so they're like terms and can combine.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Simplify 9n + 4 + 2n.

(A) 11n + 4

(B) $11n^2 + 4$

(C) 15n

(D) 92n + 4

HINT There are two n-terms (9n and 2n) and one lonely constant (4) with no partner.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Which pair are like terms?

(A) 6x and 2x

(B) 6x and 6y

(C) 6x and 6

(D) $6x$ and $2x^2$

HINT Like terms need the exact same variable raised to the exact same power.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

Simplify 5a + 3 + 2a.

(A) 7a + 3

(B) $7a^2 + 3$

(C) 10a

(D) 52a + 3

HINT Spot the like terms: 5a and 2a share the variable a. The 3 has no variable.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Mistake — Exponent Slip

1. **Expression** 4x + 3x
2. **Combine** Add the coefficients: 4 + 3 = 7
3. **Simplify** $7x^2$

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT When you add like terms, does the little exponent on x change?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What rule determines whether two terms are 'like terms'? Why can't you combine unlike terms?

SENTENCE STARTER Two terms are like terms when they have the same ___. You cannot combine ___ and ___ because they represent different ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 6.15

Simplify Algebraic Expressions

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Simplify each expression by combining like terms.

Expression	Like Terms Combined	Simplified
3a + 5a + 2	3a + 5a = 8a	
6x + 4 + 2x + 7		
10m − 3m + 5		
4y + 3 + y + 9		

SHOW YOUR WORK

2 **Write the letter of the matching item on each line.**

Match each expression to its simplified form.

_____ 8x + 3x + 1

_____ 5a − 2a + 7

_____ 4n + n + 6

_____ 12y − 5y

_____ 3m + 8 + 4m + 2

_____ 9p + 1 − 3p

(A) 11x + 1

(B) 3a + 7

(C) 5n + 6

(D) 7y

(E) 7m + 10

(F) 6p + 1

3 **Write the name of the correct group on each line.**

Simplify each expression, then sort by whether the simplified coefficient of the variable is greater than 6 or not.

Groups: Coefficient > 6 | Coefficient ≤ 6

_____ 4x + 5x

_____ 8n − 3n

_____ 3a + a + 4a

_____ 10m − 7m

_____ 2y + 5y

_____ 6p − p

4 **Decide whether each pair balances. Check Yes or No, then explain.**

Both simplify to 7x + 2 — they are equivalent.

Left side	Right side	Balanced?

EXPLAIN YOUR CHOICE

PART 2 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Spot the Mistake — Swept-Up Constant

1. **Expression** 6x + 4 + 2x
2. **Combine** Add everything: 6 + 4 + 2 = 12
3. **Simplify** 12x

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What rule determines whether two terms are 'like terms'? Why can't you combine unlike terms?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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LESSON 6.15

Simplify Algebraic Expressions

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Simplify each multi-variable expression. Only combine like terms (same variable).

Expression	Simplified
3x + 2y + 5x + y	
7a + 4b − 2a + b	
6m + 3 + 2m + n + 1	

SHOW YOUR WORK

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Simplification Error

1. **Expression** 5x + 3 + 2y + 4x
2. **Combine** 5x + 2y + 4x + 3
3. **Simplify** 11xy + 3

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

Three students simplified 4x + 3 + 2x + 5 + x differently. Student A got 7x + 8. Student B got $7x^2 + 8$. Student C got 6x + 9. Who is correct? Explain the mistakes the other two students made.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

What rule determines whether two terms are 'like terms'? Why can't you combine unlike terms?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help